

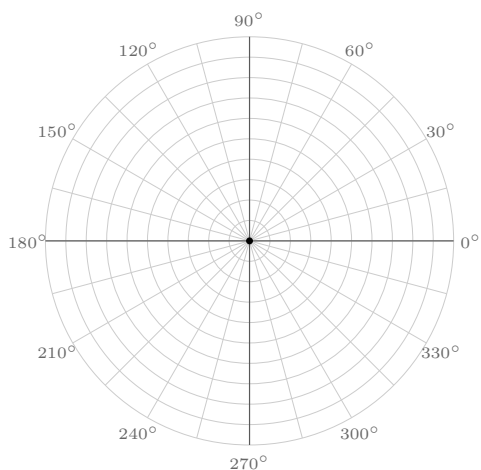
Polar Unit Test (Practice B)

*Calculator***SHOW ALL WORK FOR FULL CREDIT.****You have the entire period.**

Use your calculator to sketch a graph of each polar equation. Identify the type of polar graph it is. Show your set up (what you are putting into the calculator). (3 points each)

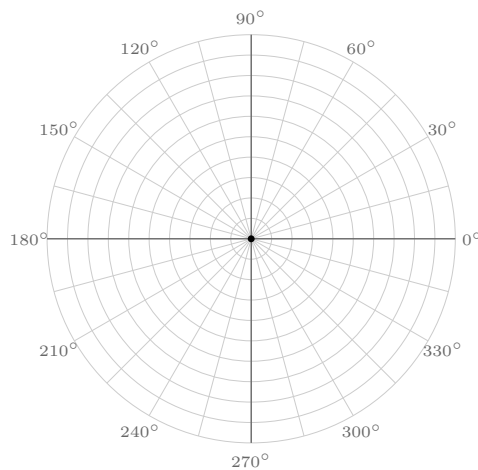
Word bank: circle cardioid limaçon rose curve lemniscate

1. $r = 5 - 5 \cos \theta$



Type: _____

2. $r = 4 \sin 3\theta$



Type: _____

Polar Unit Test (Practice B)

*Non-Calculator***SHOW ALL WORK FOR FULL CREDIT.****Simplify. (#3 is 3 points and #4 is 2 points)**

3. $\frac{\sqrt{15}}{\sqrt{-3} \cdot \sqrt{-10}}$	4. $\frac{-10 + \sqrt{-75}}{15}$
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Perform the indicated operations. Write answers in rectangular form. (2 points each for 5 & 6 and 3 points each for 7 & 8)

5. i^{146}	6. $(7 - 2i) - (-3 - 5i) - (12 + i)$
7. $\frac{\sqrt{12} - i\sqrt{12}}{\sqrt{3} + i\sqrt{3}}$	8. $-4i(2 + i)(3 - 2i)$

Solve over the set of complex numbers. (3 points)

9. $3x^2 + 2 = 2x$

Convert to trig form. (3 points)

10. $-3\sqrt{3} + 3i$

Convert to rectangular form. (3 points)

11. $4 \operatorname{cis} \frac{5\pi}{4}$

Find the product. Write it in rectangular form, using exact values. (3 points)

12. $\left[\sqrt{6} \operatorname{cis} \frac{7\pi}{12}\right] \left[\sqrt{6} \operatorname{cis} \frac{3\pi}{4}\right]$

Using $z^2 - 1, (z^2 - 1)^2 - 1, [(z^2 - 1)^2 - 1]^2 - 1, \dots$ is the following in the Julia set? Show ALL work. (3 points)

13. $z = -1$

YES / NO

Convert to trig form, then divide. Write the quotient in rectangular form. (3 points)

14. $\frac{\sqrt{6} - i\sqrt{2}}{-1 - i\sqrt{3}}$

Find the power. Write the answer in rectangular form, using exact values. (3 points)

15. $(-\sqrt{3} + i)^3$

Find all n th roots. Write your answers in polar form. (3 points)

16. $-8 + 8i\sqrt{3}, \quad n = 4$

The fractal called the Mandelbrot set: to determine if a complex number $z = a + bi$ is in this set, repeatedly compute $z, z^2 + z, (z^2 + z)^2 + z, [(z^2 + z)^2 + z]^2 + z, \dots$. The complex number z does not belong to the Mandelbrot set if any of the resulting absolute values exceeds 2. Otherwise z is in the set. Determine whether or not each number belongs to the Mandelbrot set. (3 points)

17.

a) $z = -2$

YES / NO

b) $z = 1 - i$

YES / NO

For the point given in polar coordinates, (a) state the quadrant in which the point lies if it is graphed in a rectangular coordinate system, (b) determine which is not an equivalent point, and (c) find the rectangular coordinates of the point in exact form. (3 points)

18. $(-8, \frac{3\pi}{4})$

a) $(8, \frac{7\pi}{4})$

b) $(-8, -\frac{5\pi}{4})$

c) $(8, -\frac{\pi}{4})$

d) $(8, \frac{3\pi}{4})$

Give two pairs of polar coordinates for the point, where $0^\circ \leq \theta < 360^\circ$. (3 points)

19. $(3, -3\sqrt{3})$

Convert each rectangular equation to a polar equation. (2 points each)

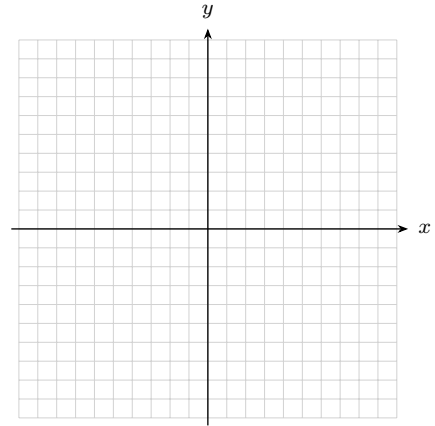
20.

a) $5x + 2y = 8$

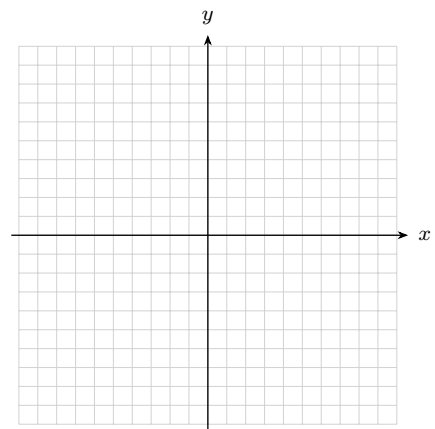
b) $x^2 + y^2 = 12$

Convert each polar equation to a rectangular equation and graph it. (No calculator) (3 points each)

21. $r = 8 \cos \theta - 6 \sin \theta$



22. $r = \frac{6}{1 - \cos \theta}$



Find the polar equation of the line. Simplify your answer. (3 points)

23. Passes through $(-3, 0^\circ)$ and $(6, 90^\circ)$

Find the polar coordinates of the points of intersection for the given interval of θ . (3 points)

24. $r = 3 + 2 \cos \theta$, $r = 3 - 2 \sin \theta$
 $0 \leq \theta < 2\pi$

Identify the type of polar graph. Use the word bank: circle, cardioid, limaçon, rose curve, lemniscate. (1 point each)

25.

a) $r = 3 \cos 4\theta$

b) $r = 6 - 3 \sin \theta$

c) $r = 5 - 5 \sin \theta$

d) $r^2 = 16 \sin 2\theta$

e) $r = 3 \sin \theta$

Honor Code (copy below & sign): On my honor, before God, I have neither given, nor received any aid on this test.

Signature

ANSWER KEY – Calculator

Left: the work, one step per line, the way it goes on paper. Right: why that line happens.

Use your calculator to sketch a graph of the polar equation. Identify the type of polar graph it is. (3 points)

1. $r = 5 - 5 \cos \theta$

WORK

Polar mode: $r1(\theta) = 5 - 5 \cos \theta$

$\theta = 0^\circ$: $r = 5 - 5 = 0$

$\theta = 90^\circ, 270^\circ$: $r = 5$

$\theta = 180^\circ$: $r = 5 - 5(-1) = 10$

$\frac{a}{b} = \frac{5}{5} = 1$

cardioid

WHY

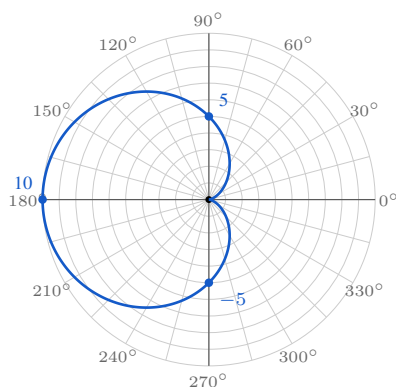
Check the angle mode if nothing shows.

Touches the pole: the dent of the heart.

Crosses the y -axis at 5 and -5 .

Farthest point, $(-10, 0)$.

Ratio exactly 1.



Use your calculator to sketch a graph of the polar equation. Identify the type of polar graph it is. (3 points)

2. $r = 4 \sin 3\theta$

WORK

Polar mode: $r1(\theta) = 4 \sin(3\theta)$

$n = 3$ (odd): 3 petals

$\theta = 30^\circ$: $r = 4 \sin 90^\circ = 4$

tips at $30^\circ, 150^\circ, 270^\circ$

rose curve

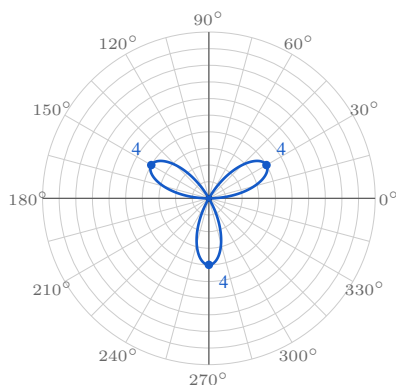
WHY

Type the parentheses around 3θ .

n odd: n petals. n even: $2n$.

Petal length = $|A| = 4$.

At 90° , $r = -4$ plots at 270° .



ANSWER KEY – Non-Calculator

Left: the work, one step per line, the way it goes on paper. Right: why that line happens.

Simplify. (3 points)

3. $\frac{\sqrt{15}}{\sqrt{-3} \cdot \sqrt{-10}}$

WORK

$$\begin{aligned}\frac{\sqrt{15}}{i\sqrt{3} \cdot i\sqrt{10}} &= \frac{\sqrt{15}}{i^2\sqrt{30}} \\ &= \frac{\sqrt{15}}{-\sqrt{30}} = -\sqrt{\frac{15}{30}} = -\frac{1}{\sqrt{2}} \\ &= -\frac{1}{\sqrt{2}} \cdot \frac{\sqrt{2}}{\sqrt{2}} = \boxed{-\frac{\sqrt{2}}{2}}\end{aligned}$$

WHY

Pull out i first, then multiply.

$i^2 = -1$; one radical, reduce inside.

Rationalize.

Trap: Rationalize by multiplying by the radical over itself, not by a whole number.

Simplify. (2 points)

4. $\frac{-10 + \sqrt{-75}}{15}$

WORK

$$\begin{aligned}\frac{-10 + 5i\sqrt{3}}{15} \\ &= -\frac{10}{15} + \frac{5\sqrt{3}}{15}i = \boxed{-\frac{2}{3} + \frac{\sqrt{3}}{3}i}\end{aligned}$$

WHY

$\sqrt{-75} = i\sqrt{25 \cdot 3} = 5i\sqrt{3}$.

Split, then reduce each fraction by 5.

Trap: Reduce both parts by the same factor; the i part keeps its own coefficient.

Perform the indicated operation. Write the answer in rectangular form. (2 points)

5. i^{146}

WORK

$$\begin{aligned}146 \div 4 &= 36 \text{ R } 2 \\ i^{146} &= i^2 = \boxed{-1 + 0i}\end{aligned}$$

WHY

Only the remainder matters.

Perform the indicated operations. Write the answer in rectangular form. (2 points)

6. $(7 - 2i) - (-3 - 5i) - (12 + i)$

WORK

$$\begin{aligned}7 - 2i + 3 + 5i - 12 - i \\ \text{real: } 7 + 3 - 12 = -2; \text{ imaginary: } -2 + 5 - 1 = 2 \\ \boxed{-2 + 2i}\end{aligned}$$

WHY

Distribute each minus sign to both parts.

Trap: $-(9 + 5i)$ is $-9 - 5i$: both signs flip.

Perform the indicated operation. Write the answer in rectangular form. (3 points)

7. $\frac{\sqrt{12} - i\sqrt{12}}{\sqrt{3} + i\sqrt{3}}$

WORK

$$\frac{\sqrt{12} - i\sqrt{12}}{\sqrt{3} + i\sqrt{3}} \cdot \frac{\sqrt{3} - i\sqrt{3}}{\sqrt{3} - i\sqrt{3}}$$

top: $\sqrt{36} - i\sqrt{36} - i\sqrt{36} + i^2\sqrt{36} = 6 - 12i - 6 = -12i$
 bottom: $3 + 3 = 6$
 $\frac{-12i}{6} = \boxed{0 - 2i}$

WHY

Multiply by the conjugate of the bottom.

$$\sqrt{12}\sqrt{3} = \sqrt{36} = 6.$$

$$a^2 + b^2.$$

Perform the indicated operations. Write the answer in rectangular form. (3 points)

8. $-4i(2 + i)(3 - 2i)$

WORK

$$(-8i - 4i^2)(3 - 2i) = (4 - 8i)(3 - 2i)$$

$$= 12 - 8i - 24i + 16i^2$$

$$= 12 - 32i - 16 = \boxed{-4 - 32i}$$

WHY

Distribute the $-4i$ first; $-4i^2 = +4$.
FOIL.

Solve the equation over the set of complex numbers. (3 points)

9. $3x^2 + 2 = 2x$

WORK

$$3x^2 - 2x + 2 = 0$$

$$x = \frac{2 \pm \sqrt{4 - 4(3)(2)}}{2(3)} = \frac{2 \pm \sqrt{-20}}{6}$$

$$= \frac{2 \pm 2i\sqrt{5}}{6}$$

$$\boxed{x = \frac{1}{3} \pm \frac{\sqrt{5}}{3}i}$$

WHY

Zero on one side first.

$$a = 3, b = -2, c = 2; -b = +2.$$

$$\sqrt{-20} = 2i\sqrt{5}.$$

Reduce both parts.

Convert to trig form. (3 points)

10. $-3\sqrt{3} + 3i$

WORK

$$r = \sqrt{27 + 9} = 6$$

$$\theta = \tan^{-1}\left(\frac{3}{-3\sqrt{3}}\right) = \tan^{-1}\left(-\frac{1}{\sqrt{3}}\right) = -30^\circ$$

$$\theta = -30^\circ + 180^\circ = 150^\circ$$

$$\boxed{6 \text{ cis } 150^\circ}$$

WHY

$$r = \sqrt{x^2 + y^2}.$$

$\tan^{-1}$ only gives QI or QIV.

$x < 0, y > 0$ is QII: add 180° .

Convert to rectangular form. (3 points)

11. $4 \text{ cis } \frac{5\pi}{4}$

WORK

$$4\left(\cos \frac{5\pi}{4} + i \sin \frac{5\pi}{4}\right) = 4\left(-\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}i\right)$$

$$= \boxed{-2\sqrt{2} - 2\sqrt{2}i}$$

WHY

$\frac{5\pi}{4}$ is QIII: both negative.

Distribute the 4.

Find the product. Write the product in rectangular form, using exact values. (3 points)

12. $[\sqrt{6}(\cos \frac{7\pi}{12} + i \sin \frac{7\pi}{12})][\sqrt{6}(\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4})]$

WORK

$$\sqrt{6} \cdot \sqrt{6} \operatorname{cis} \left(\frac{7\pi}{12} + \frac{9\pi}{12} \right)$$

$$= 6 \operatorname{cis} \frac{16\pi}{12} = 6 \operatorname{cis} \frac{4\pi}{3}$$

$$= 6 \left(-\frac{1}{2} - \frac{\sqrt{3}}{2}i \right)$$

$$= \boxed{-3 - 3\sqrt{3}i}$$

WHY

Product theorem. Common denominator:

$$\frac{3\pi}{4} = \frac{9\pi}{12}.$$

Reduce the fraction.

$$\frac{4\pi}{3} \text{ is QIII.}$$

Using $z^2 - 1$, $(z^2 - 1)^2 - 1$, $[(z^2 - 1)^2 - 1]^2 - 1, \dots$ is the following in the Julia set? Show ALL work. (3 points)

13. $z = -1$

YES / NO

WORK

$$(-1)^2 - 1 = 0$$

$$0^2 - 1 = -1, (-1)^2 - 1 = 0, \dots$$

$$\text{never exceeds 2: } \boxed{\text{YES}}$$

WHY

$$\text{First step; } |0| = 0.$$

$$\text{Cycles } 0, -1, 0, -1.$$

Convert to trigonometric form, then divide. Write the quotient in rectangular form, using exact values. (3 points)

14. $\frac{\sqrt{6} - i\sqrt{2}}{-1 - i\sqrt{3}}$

WORK

$$r_1 = \sqrt{6 + 2} = 2\sqrt{2}$$

$$\theta_1 = \tan^{-1} \left(\frac{-\sqrt{2}}{\sqrt{6}} \right) = \tan^{-1} \left(-\frac{1}{\sqrt{3}} \right) = -30^\circ \rightarrow 330^\circ$$

$$r_2 = \sqrt{1 + 3} = 2, \theta_2 = \tan^{-1}(\sqrt{3}) = 60^\circ$$

$$\theta_2 = 60^\circ + 180^\circ = 240^\circ$$

$$\frac{2\sqrt{2} \operatorname{cis} 330^\circ}{2 \operatorname{cis} 240^\circ} = \sqrt{2} \operatorname{cis} 90^\circ$$

$$= \sqrt{2}(0 + 1i) = \boxed{\sqrt{2}i}$$

WHY

Top first.

$$\frac{\sqrt{2}}{\sqrt{6}} = \frac{1}{\sqrt{3}}. \text{ QIV: add } 360^\circ.$$

Reference angle.

$x < 0, y < 0$ is QIII.

Quotient theorem.

$$\cos 90^\circ = 0, \sin 90^\circ = 1.$$

Find the power. Write the answer in rectangular form, using exact values. (3 points)

15. $(-\sqrt{3} + i)^3$

WORK

$$r = \sqrt{3 + 1} = 2$$

$$\theta = \tan^{-1} \left(-\frac{1}{\sqrt{3}} \right) = -30^\circ, -30^\circ + 180^\circ = 150^\circ$$

$$(2 \operatorname{cis} 150^\circ)^3 = 8 \operatorname{cis} 450^\circ$$

$$= 8 \operatorname{cis} 90^\circ = \boxed{8i}$$

WHY

$x < 0, y > 0$ is QII.

De Moivre.

$$450^\circ - 360^\circ = 90^\circ.$$

Find all n th roots. Write your answers in polar form. (3 points)

16. $-8 + 8i\sqrt{3}, \quad n = 4$

WORK

$$r = \sqrt{64 + 192} = \sqrt{256} = 16$$

$$\theta = \tan^{-1}(-\sqrt{3}) = -60^\circ, -60^\circ + 180^\circ = 120^\circ$$

$$\sqrt[4]{16} = 2$$

$$\alpha = \frac{120^\circ + 360^\circ k}{4} = 30^\circ + 90^\circ k$$

$$\boxed{2 \operatorname{cis} 30^\circ, 2 \operatorname{cis} 120^\circ, 2 \operatorname{cis} 210^\circ, 2 \operatorname{cis} 300^\circ}$$

WHY

$$(8\sqrt{3})^2 = 64 \cdot 3 = 192.$$

$x < 0, y > 0$ is QII.

n th root theorem: root of r .

$$k = 0, 1, 2, 3: \text{ four roots, } 90^\circ \text{ apart.}$$

Polar form was asked, so stop here.

The fractal called the **Mandelbrot set**: to determine if a complex number $z = a + bi$ is in this set, repeatedly compute $z, z^2 + z, (z^2 + z)^2 + z, [(z^2 + z)^2 + z]^2 + z, \dots$. The complex number z does **not** belong to the Mandelbrot set if any of the resulting absolute values exceeds 2. Otherwise z is in the set. Determine whether or not each number belongs to the Mandelbrot set. (3 points)

17.

WORK

a) $-2, (-2)^2 + (-2) = 2, 2^2 + (-2) = 2$

YES

b) $(1 - i)^2 + (1 - i) = -2i + 1 - i = 1 - 3i$

$|1 - 3i| = \sqrt{1 + 9} = \sqrt{10} \approx 3.16 > 2$: **NO**

WHY

Stuck at 2 forever.

$|2| = 2$ does not exceed 2.

$(1 - i)^2 = 1 - 2i + i^2 = -2i$.

For the point given in polar coordinates, (a) state the quadrant in which the point lies if it is graphed in a rectangular coordinate system, (b) determine which is not an equivalent point, and (c) find the rectangular coordinates of the point in exact form. (3 points)

18. $(-8, \frac{3\pi}{4})$

WORK

(a) face $\frac{3\pi}{4}$ (QII), walk backward: **QIV**

(b) $(-8, \frac{3\pi}{4}) = (8, \frac{3\pi}{4} + \pi) = (8, \frac{7\pi}{4})$

$(8, -\frac{\pi}{4})$ same as $\frac{7\pi}{4}$; $-\frac{5\pi}{4}$ same as $\frac{3\pi}{4}$

$(8, \frac{3\pi}{4})$ is in QII: **d**

(c) $x = -8 \cos \frac{3\pi}{4} = -8 \left(-\frac{\sqrt{2}}{2}\right) = 4\sqrt{2}$

$y = -8 \sin \frac{3\pi}{4} = -8 \left(\frac{\sqrt{2}}{2}\right) = -4\sqrt{2}$

$(4\sqrt{2}, -4\sqrt{2})$

WHY

Negative r lands in the opposite quadrant.

$(-r, \theta) = (r, \theta \pm \pi)$.

$\pm 2\pi$ keeps the point.

Keep r 's sign.

QIV, matches (a).

Give two pairs of polar coordinates for the point, where $0^\circ \leq \theta < 360^\circ$. (3 points)

19. $(3, -3\sqrt{3})$

WORK

$r = \sqrt{9 + 27} = 6$

$\tan \theta = \frac{-3\sqrt{3}}{3} = -\sqrt{3}, \theta = -60^\circ$

$\theta = -60^\circ + 360^\circ = 300^\circ$

$(6, 300^\circ)$, $(-6, 120^\circ)$

WHY

$r^2 = x^2 + y^2$.

$\tan^{-1}$ answer.

$x > 0, y < 0$: QIV.

Second pair: $(-r, \theta - 180^\circ)$.

Convert each rectangular equation to a polar equation. (2 points each)

20.

WORK

a) $5r \cos \theta + 2r \sin \theta = 8$

$r = \frac{8}{5 \cos \theta + 2 \sin \theta}$

b) $r^2(\cos^2 \theta + \sin^2 \theta) = 12, r^2 = 12$

$r = \pm\sqrt{12} = \pm 2\sqrt{3}$

$r = 2\sqrt{3}$ or $r = -2\sqrt{3}$

WHY

Substitute.

Factor r , divide.

Pythagorean identity.

$\pm$, then simplify.

Convert each polar equation to a rectangular equation and graph it. (No calculator) (3 points each)

21. $r = 8 \cos \theta - 6 \sin \theta$

WORK

$$r^2 = 8r \cos \theta - 6r \sin \theta$$

$$x^2 + y^2 = 8x - 6y$$

$$x^2 - 8x + 16 + y^2 + 6y + 9 = 16 + 9$$

$$(x - 4)^2 + (y + 3)^2 = 25$$

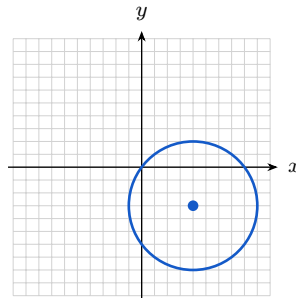
WHY

Multiply both sides by r .

Substitute.

Complete the square twice.

Circle, center $(4, -3)$, radius 5.



Convert each polar equation to a rectangular equation and graph it. (No calculator) (3 points each)

22. $r = \frac{6}{1 - \cos \theta}$

WORK

$$r(1 - \cos \theta) = 6, \quad r - r \cos \theta = 6$$

$$r - x = 6, \quad r = x + 6$$

$$x^2 + y^2 = x^2 + 12x + 36$$

$$y^2 = 12x + 36$$

$$y^2 = 12(x + 3)$$

WHY

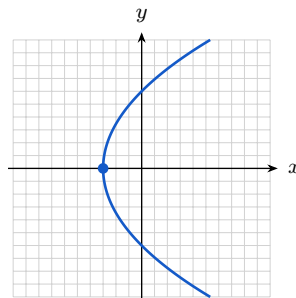
Clear the fraction.

$r \cos \theta = x$; isolate r .

Square both sides; $r^2 = x^2 + y^2$.

x^2 cancels.

Sideways parabola, vertex $(-3, 0)$, opens right.



Find the polar equation of the line. Simplify your answer. (3 points)

23. Passes through $(-3, 0^\circ)$ and $(6, 90^\circ)$

WORK

$$(-3, 0^\circ): x = -3 \cos 0^\circ = -3, y = 0, (-3, 0)$$

$$(6, 90^\circ): x = 0, y = 6, (0, 6)$$

$$m = \frac{6 - 0}{0 - (-3)} = 2, \quad y = 2x + 6$$

$$2x - y = -6$$

$$2r \cos \theta - r \sin \theta = -6$$

$$r = \frac{-6}{2 \cos \theta - \sin \theta}$$

WHY

Negative r puts it on the left.

Slope, y -intercept 6.

Standard form.

Substitute.

Factor r , divide.

Find the polar coordinates of the points of intersection for the given interval of θ . (3 points)

24. $r = 3 + 2 \cos \theta$, $r = 3 - 2 \sin \theta$, $0 \leq \theta < 2\pi$

WORK

$$3 + 2 \cos \theta = 3 - 2 \sin \theta$$

$$\sin \theta = -\cos \theta$$

$$\theta = \frac{3\pi}{4}, \frac{7\pi}{4}$$

$$\theta = \frac{3\pi}{4}: r = 3 + 2 \left(-\frac{\sqrt{2}}{2} \right) = 3 - \sqrt{2}$$

$$\theta = \frac{7\pi}{4}: r = 3 + 2 \left(\frac{\sqrt{2}}{2} \right) = 3 + \sqrt{2}$$

$$\left(3 - \sqrt{2}, \frac{3\pi}{4} \right), \left(3 + \sqrt{2}, \frac{7\pi}{4} \right)$$

WHY

Set $r = r$.

Do not divide by $\cos \theta$.

Unit circle points where $y = -x$.

Plug back in.

Neither curve reaches the pole ($r \geq 1$).

Identify the type of polar graph. Use the word bank: circle, cardioid, limaçon, rose curve, lemniscate. (1 point each)

25.

WORK

a)

b)

c)

d)

e)

WHY

$n = 4$ even: $2n = 8$ petals, each 3 long.

$$\frac{a}{b} = \frac{6}{3} = 2 \neq 1.$$

$$\frac{a}{b} = \frac{5}{5} = 1.$$

r^2 equation.

$$r^2 = 3r \sin \theta \text{ gives } x^2 + y^2 = 3y.$$